📘 UNIT 1 — INTRODUCTION TO MACHINE LEARNING

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1. INTRODUCTION TO MACHINE LEARNING

1.1 What is Machine Learning?

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to **learn patterns from data and make decisions or predictions without being explicitly programmed for every situation.**

Simple Definition

> **Machine Learning = Learning from data to make predictions or decisions.**

Example

Suppose we want to identify spam emails.

Traditional Programming

```
Rules + Email
     ↓
  Program
     ↓
Spam / Not Spam
```

We manually create rules such as:

```
IF email contains "FREE MONEY"
    → Spam
```

But spam emails can have millions of variations.

Machine Learning

```
Past Emails + Labels
        ↓
   ML Algorithm
        ↓
```

```
      Model
        ↓
New Email → Spam / Not Spam
```

The computer learns useful patterns from previous examples.

1.2 Formal Definition of Machine Learning

A commonly used definition by **Tom Mitchell** is:

> "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E."

In simple terms:

```
Experience → Learning → Better Performance
```

For example:

Spam Detection

- **E (Experience):** Previous emails
- **T (Task):** Classify emails
- **P (Performance):** Classification accuracy

If the system becomes better at detecting spam after seeing more emails, it is learning.

1.3 Traditional Programming vs Machine Learning

Traditional Programming

```
        DATA
          +
        RULES
          ↓
       PROGRAM
          ↓
       OUTPUT
```

Machine Learning

```
      DATA
        +
     OUTPUT
        ↓
ML ALGORITHM
        ↓
      MODEL
        ↓
     PREDICTION
```

Difference

Traditional Programming	Machine Learning
Rules are explicitly written	Rules/patterns are learned
Programmer defines logic	Algorithm learns logic
Good for well-defined rules	Good for complex patterns
Data + Rules → Output	Data + Output → Model
Less adaptive	Can improve with data

1.4 Why Machine Learning?

Traditional programming becomes difficult when:

- Rules are extremely complex
- Rules continuously change
- Data is very large
- Patterns are difficult for humans to define
- The problem involves images, speech, language, etc.

Example: Face Recognition

It is difficult to manually write rules for:

```
Eyes
Nose
Mouth
Face shape
```

```
Skin
Lighting
Angle
Expression
```

Instead, ML can learn these patterns from thousands/millions of images.

1.5 Real-Life Applications of Machine Learning

Healthcare

- Disease prediction
- Medical image analysis
- Drug discovery
- Patient risk prediction

Finance

- Fraud detection
- Credit scoring
- Stock analysis
- Risk assessment

E-Commerce

- Product recommendation
- Customer segmentation
- Price prediction

Social Media

- Content recommendation
- Face recognition
- Spam detection
- Personalized feeds

Transportation

- Self-driving vehicles
- Traffic prediction
- Route optimization

Cybersecurity

- Malware detection
- Intrusion detection

- Anomaly detection

Natural Language Processing

- Chatbots
- Translation
- Sentiment analysis
- Speech recognition

---

2. DATA

2.1 What is Data?

Data is a collection of facts, observations, measurements, or information that can be processed to obtain useful knowledge.

Example:

```
Age = 21
Salary = ₹50,000
Height = 175 cm
City = Pune
```

Together these values represent information about an individual.

2.2 Dataset

A **dataset** is a collection of related data used for analysis or machine learning.

Example:

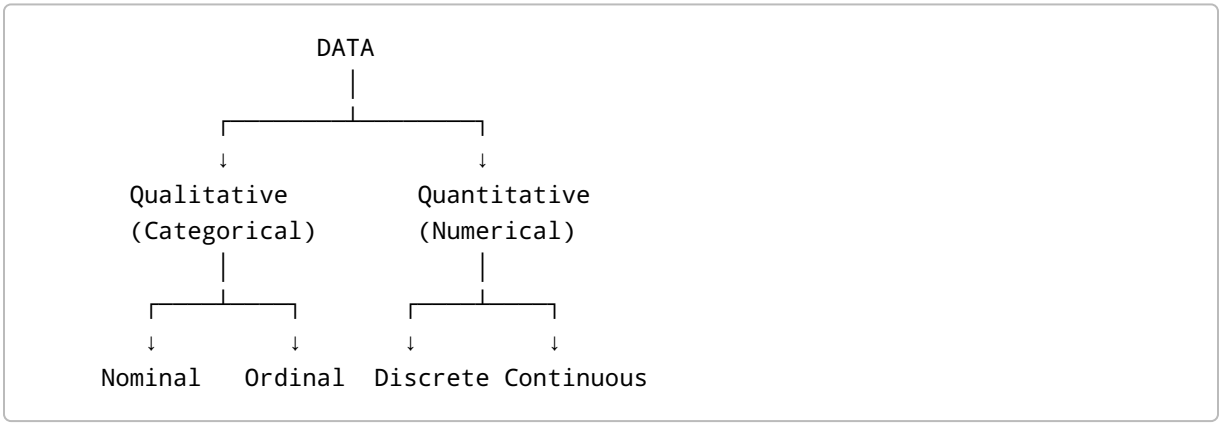
Age	Salary	Experience	Purchased
22	30000	1	No
25	45000	3	Yes
30	70000	7	Yes
21	25000	0	No

Here:

- Rows → observations/samples
- Columns → features/attributes
- Purchased → target/label

2.3 Types of Data

Data can broadly be divided into:



2.4 Qualitative / Categorical Data

Categorical data represents **categories or groups**.

Examples:

```
Gender = Male / Female
Color = Red / Blue / Green
City = Pune / Delhi / Mumbai
Blood Group = A / B / AB / O
```

It generally represents qualities rather than numerical quantities.

2.5 Quantitative / Numerical Data

Numerical data represents quantities that can be measured or counted.

Examples:

```
Age = 20
Height = 175 cm
Salary = ₹50,000
Number of children = 2
```

It is divided into:

Discrete Data

Countable values.

Examples:

```
Number of students = 50
Number of cars = 10
Number of children = 2
```

Usually represented by integers.

Continuous Data

Can take any value within a range.

Examples:

```
Height = 175.52 cm
Weight = 68.73 kg
Temperature = 27.63°C
```

---

3. SCALES OF MEASUREMENT

The four major scales are:

```
1. Nominal
2. Ordinal
3. Interval
4. Ratio
```

A useful memory trick:

> **NOIR → Nominal, Ordinal, Interval, Ratio**

3.1 Nominal Scale

Nominal data represents **names or categories without any inherent order**.

Examples:

```
Gender
Blood group
Color
Country
City
```

Example:

```
Color:
Red
Blue
Green
```

There is no meaningful:

```
Red < Blue < Green
```

Characteristics

- Categories only
- No ranking
- No meaningful arithmetic operations

3.2 Ordinal Scale

Ordinal data has **categories with an order/ranking**, but the difference between categories is not necessarily equal.

Example:

```
Poor < Average < Good < Excellent
```

Another example:

```
1 Star < 2 Star < 3 Star < 4 Star < 5 Star
```

We know:

```
Excellent > Good
```

But we cannot say that the difference between Poor and Average is exactly equal to the difference between Good and Excellent.

Characteristics

- Categories
- Ranking exists
- Differences are not necessarily meaningful

3.3 Interval Scale

Interval data has:

- Ordered values
- Equal intervals
- No true/absolute zero

Example:

Temperature in Celsius

```
10°C
20°C
30°C
```

Difference between:

```
10°C and 20°C = 10°C
20°C and 30°C = 10°C
```

But:

```
20°C ≠ twice as hot as 10°C
```

because 0°C does not represent complete absence of temperature.

Examples

- Celsius temperature
- Fahrenheit temperature
- Calendar years

3.4 Ratio Scale

Ratio scale contains:

- Order
- Equal intervals
- True zero
- Meaningful ratios

Examples:

```
Height
Weight
Age
Distance
Income
Time
```

Example:

```
20 kg is twice 10 kg
```

This statement is meaningful because zero weight represents the absence of weight.

3.5 Comparison of Measurement Scales

Property	Nominal	Ordinal	Interval	Ratio
Categories	✓	✓	Sometimes	Sometimes
Order	✗	✓	✓	✓
Equal intervals	✗	✗	✓	✓
True zero	✗	✗	✗	✓
Addition meaningful	✗	Limited	✓	✓
Ratios meaningful	✗	✗	✗	✓

---

4. DATA, FEATURES AND PATTERNS

4.1 Sample / Instance / Observation

A single record in a dataset is called an:

- Instance
- Sample
- Observation
- Data point

Example:

Age	Salary	Experience	Purchased
25	45000	3	Yes

This entire row represents one observation.

4.2 Feature

A **feature** is an individual measurable property or attribute used to describe an observation.

Example:

```
Age
Salary
Height
Weight
Experience
```

For:

```
Age = 25
Salary = 45000
Experience = 3
```

the features are:

```
X₁ = Age
X₂ = Salary
X₃ = Experience
```

4.3 Feature Vector

The values of all features for one observation form a **feature vector**.

Example:

```
Age = 25
Salary = 45000
Experience = 3
```

Feature vector:

$$X = [25,\ 45000,\ 3]$$

4.4 Label / Target

The output that we want the ML model to predict is called:

- Label
- Target

- Output variable
- Dependent variable

Example:

Age	Salary	Experience	Purchased
25	45000	3	Yes

Here:

```
Features = Age, Salary, Experience
Target = Purchased
```

4.5 Pattern

A **pattern** is a meaningful relationship, regularity, or structure present in data.

Example:

```
More experience
       ↓
Usually higher salary
```

This relationship can be a pattern.

Another example:

```
Customers who frequently buy X
       ↓
Often buy Y
```

ML algorithms attempt to discover useful patterns from data.

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5. LEARNING TASKS

Learning tasks can broadly be divided into:



5.1 Descriptive Tasks

The objective is to **understand, summarize, or discover structure in existing data.**

Question:

> "What is happening in the data?"

Examples:

Customer Segmentation

```
Customer data
      ↓
Find natural groups
      ↓
Group 1
Group 2
Group 3
```

Other examples

- Clustering
- Association rule discovery
- Data summarization
- Pattern discovery
- Anomaly detection
- Visualization

5.2 Predictive Tasks

The objective is to use existing data to **predict an unknown/future outcome.**

Question:

"What will happen for new data?"

Examples:

```
House features → House price
Email → Spam / Not Spam
Student information → Pass / Fail
```

Predictive tasks include:

Classification

Predict a category.

Regression

Predict a numerical value.

5.3 Descriptive vs Predictive

Descriptive	Predictive
Understand existing data	Predict unknown outcomes
Finds patterns/structure	Makes predictions
Often unsupervised	Often supervised
Clustering	Classification
Association	Regression

---

6. LEARNING PARADIGMS

The three major learning paradigms are:

```
Machine Learning
│
├── Supervised Learning
├── Unsupervised Learning
└── Reinforcement Learning
```

7. SUPERVISED LEARNING

In **supervised learning**, the model learns from **labeled data**.

That means we already know the desired output for training examples.

```
Input + Correct Output
        ↓
   Learning Algorithm
        ↓
       Model
        ↓
Prediction for new input
```

Example:

Hours Studied	Attendance	Result
2	60%	Fail
5	80%	Pass
8	95%	Pass

The model learns:

```
Hours + Attendance
        ↓
      Result
```

7.1 Types of Supervised Learning

```
Supervised Learning
        │
    ┌───┴────┐
    ↓        ↓
Classification Regression
```

7.2 Classification

Classification predicts a **discrete/categorical class**.

Examples:

```
Spam / Not Spam
Pass / Fail
Cat / Dog
Fraud / Not Fraud
Positive / Negative
```

Example:

```
Email
  ↓
Classifier
  ↓
Spam
```

7.3 Regression

Regression predicts a **continuous numerical value**.

Examples:

```
House → ₹50 lakh
Student → 82.5 marks
Car → ₹8.2 lakh
Temperature → 31.5°C
```

Example:

```
House size + Location + Rooms
              ↓
         Regression Model
              ↓
       ₹75,00,000
```

8. UNSUPERVISED LEARNING

In unsupervised learning, the data **does not contain target labels**.

```
Input Data
    ↓
ML Algorithm
    ↓
Discover hidden structure
```

Example:

```
Customer Data
      ↓
  Clustering
      ↓
┌─────┬─────┬─────┐
│Group│Group│Group│
│  A  │  B  │  C  │
└─────┴─────┴─────┘
```

The algorithm discovers groups automatically.

8.1 Applications of Unsupervised Learning

- Customer segmentation
- Document clustering
- Image grouping
- Anomaly detection
- Market basket analysis
- Dimensionality reduction

---

9. REINFORCEMENT LEARNING

Reinforcement Learning (RL) is a learning paradigm where an **agent learns by interacting with an environment and receiving rewards or penalties.**

Main components:

```
Agent
Environment
State
Action
Reward
Policy
```

9.1 Reinforcement Learning Process



Example: Chess-playing AI

```
Current Board
      ↓
     Agent
      ↓
 Select Move
      ↓
   Game State
      ↓
 Reward / Penalty
      ↓
 Learn Better Strategy
```

9.2 Supervised vs Unsupervised vs Reinforcement

Property	Supervised	Unsupervised	Reinforcement
Labels	Yes	No	No fixed labels
Feedback	Correct answer	No direct answer	Reward/penalty
Goal	Predict output	Discover structure	Maximize reward
Example	Spam detection	Customer clustering	Game AI
Learning style	From examples	From patterns	Trial and error

---

10. LEARNING MODELS

A **model** is a learned mathematical or computational representation of patterns in data.

Example:

Suppose training data indicates:

```
More experience → Higher salary
```

A regression algorithm might learn:

$$Salary = 5000 + 5000(Experience)$$

This learned relationship is the **model**.

10.1 Algorithm vs Model

This distinction is very important.

Algorithm

The procedure used to learn.

Example:

```
Linear Regression Algorithm
```

Model

The result obtained after training.

Example:

$$y = 5000 + 5000x$$

Therefore:

```
Algorithm
    ↓
Training Data
    ↓
Learning
    ↓
Learned Model
    ↓
Prediction
```

10.2 Training

Training is the process of learning patterns/parameters from training data.

```
Training Data
     ↓
ML Algorithm
     ↓
Learn Parameters
     ↓
Model
```

10.3 Testing

After training, the model is evaluated on unseen data.

```
Training Data → Learn
Testing Data  → Evaluate
```

The testing data should represent data the model has not previously seen.

10.4 Generalization

Generalization is the ability of a trained model to perform well on unseen data.

A good ML model should not simply memorize training examples.

It should learn general patterns.

```
Training Data
     ↓
Learn general pattern
     ↓
Unseen Data
     ↓
Good Prediction
```

---

11. DATA AND DIMENSIONALITY

11.1 What is Dimensionality?

In ML, dimensionality generally refers to the **number of features/variables used to represent each data point.**

Example:

```
Age
Salary
Experience
```

There are 3 features.

Therefore:

$$Dimension = 3$$

Feature vector:

$$X = [Age, Salary, Experience]$$

has 3 dimensions.

11.2 High-Dimensional Data

Suppose a dataset contains:

```
10 features → Low dimensional
100 features → Higher dimensional
10,000 features → High dimensional
```

Examples:

Image

A 100 × 100 grayscale image:

$$100 \times 100 = 10,000$$

pixels/features.

Text

A document represented using thousands of words may have thousands of dimensions.

11.3 Problems with High Dimensionality

High dimensionality can cause:

1. Computational complexity

More features require more computation.

2. Memory requirements

More features require more storage.

3. Redundancy

Some features may contain similar information.

4. Noise

Irrelevant features can introduce noise.

5. Overfitting

A model may learn noise instead of useful patterns.

6. Curse of Dimensionality

As dimensionality increases, data becomes increasingly sparse and many ML techniques become less effective.

---

12. FEATURE SETS

A **feature set** is the collection of features used to represent data.

Example:

```
Student Dataset

Age
Attendance
Study Hours
Previous Marks
```

Feature set:

$$F = \{Age, Attendance, StudyHours, PreviousMarks\}$$

For one student:

$$X = [20, 85, 6, 78]$$

12.1 Feature Matrix

A dataset can be represented as a matrix.

Example:

$$X = \begin{bmatrix} 20 & 85 & 6 & 78 \\ 21 & 90 & 8 & 85 \\ 19 & 75 & 4 & 65 \end{bmatrix}$$

Here:

```
Rows = samples
Columns = features
```

Therefore:

$$3\ samples \times 4\ features$$

---

13. FEATURE EXTRACTION

Feature extraction is the process of transforming raw data into a set of useful features.

```
Raw Data
   ↓
Feature Extraction
   ↓
Useful Representation
```

Example 1 — Image

Raw image:

```
Pixels
 ↓
Edges
 ↓
Shapes
 ↓
Textures
 ↓
Useful Features
```

Example 2 — Text

Raw text:

```
"I love this movie"
```

can be converted into numerical features using techniques such as:

- Bag of Words
- TF-IDF
- Word embeddings

Example 3 — Audio

Raw audio:

```
Audio Signal
     ↓
Frequency Features
     ↓
MFCC / Spectral Features
```

---

14. FEATURE SUBSET SELECTION

Feature selection means selecting a **subset of existing features** that are useful for the ML task.

Suppose:

```
Original Features:

Age
Salary
Height
Weight
Favorite Color
Employee ID
```

After feature selection:

```
Age
Salary
Height
Weight
```

We have removed irrelevant/redundant features.

14.1 Why Feature Selection?

Advantages:

- Reduces dimensionality
- Reduces computation
- Reduces noise
- Can improve model performance
- Reduces overfitting
- Makes models easier to interpret

14.2 Feature Selection vs Feature Extraction

Feature Selection	Feature Extraction
Selects existing features	Creates/transforms features
Original features remain	New representation may be created
Removes irrelevant features	Compresses information
Example: select Age + Salary	Example: PCA components

Memory Trick

Selection = Choose

Extraction = Create

---

15. FEATURE TRANSFORMATION

Feature transformation means **changing the representation or scale of features**.

Examples:

- Normalization
- Standardization
- Log transformation
- Encoding
- Mathematical transformations

15.1 Normalization

Normalization commonly scales values into a fixed range such as:

$$[0, 1]$$

Min-Max normalization:

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Example:

If:

$$x = 50, \quad x_{min} = 0, \quad x_{max} = 100$$

then:

$$x' = \frac{50 - 0}{100 - 0} = 0.5$$

15.2 Standardization

Standardization transforms data so that it has approximately:

```
Mean = 0
Standard deviation = 1
```

Formula:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x = original value
- μ = mean
- σ = standard deviation

15.3 Why Feature Transformation?

Suppose:

```
Age        = 20–60
Salary     = 20,000–2,00,000
Experience = 0–20
```

Salary has a much larger numerical scale.

Some algorithms can be affected by this difference.

Scaling puts features on comparable scales.

---

16. DIMENSIONALITY REDUCTION

Dimensionality reduction is the process of **reducing the number of features while preserving important information.**

Example:

```
100 features
     ↓
Dimensionality Reduction
     ↓
10 features
```

16.1 Why Dimensionality Reduction?

Benefits

1. Reduces computation

2. Reduces memory usage
3. Removes redundancy
4. Reduces noise
5. Helps visualization
6. Can reduce overfitting
7. Makes models simpler

16.2 Two Major Approaches



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17. PCA — PRINCIPAL COMPONENT ANALYSIS

17.1 Definition

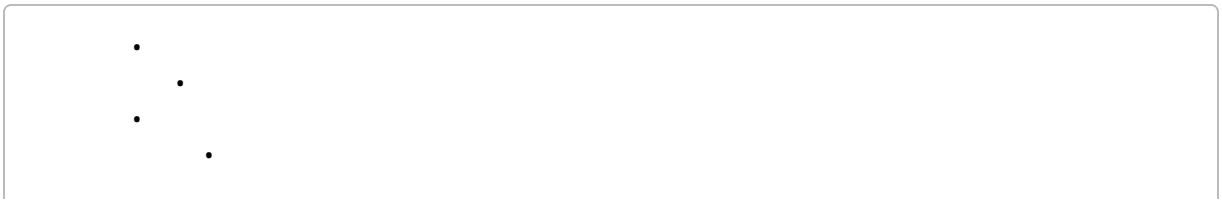
Principal Component Analysis (PCA) is an unsupervised dimensionality reduction technique that transforms the original features into a smaller set of new variables called **principal components**, while retaining as much variance as possible.

Main idea:

> **PCA finds directions in the data along which the data varies the most.**

17.2 Intuition Behind PCA

Suppose data looks like:





The points have a major direction of variation.

PCA finds:



PC1 captures the maximum variance.

Then:



PC2 captures the maximum remaining variance while being perpendicular to PC1.

17.3 Principal Components

Principal components are new axes/directions created from the original features.

For example:



Instead of keeping 4 original dimensions, we might retain only 2.

17.4 Important PCA Terms

Variance

Measures how spread out the data is.

Covariance

Measures how two variables change together.

Eigenvector

Represents a direction.

Eigenvalue

Represents the amount of variance/information associated with that direction.

Principal Component

A direction corresponding to high variance.

17.5 PCA Procedure

The general PCA algorithm:

```
Step 1 → Collect data
Step 2 → Center/standardize data
Step 3 → Calculate covariance matrix
Step 4 → Calculate eigenvalues/eigenvectors
Step 5 → Sort by eigenvalues
Step 6 → Select top components
Step 7 → Project data onto components
```

17.6 PCA Mathematical Process

Given a dataset:

$$X$$

Step 1: Standardize

$$Z = \frac{X - \mu}{\sigma}$$

Step 2: Calculate covariance matrix

$$C = \frac{1}{n-1} Z^T Z$$

Step 3: Calculate eigenvalues and eigenvectors

$$Cv = \lambda v$$

where:

- C = covariance matrix
- v = eigenvector
- λ = eigenvalue

Step 4: Sort eigenvalues

Largest eigenvalue:

$$\lambda_1$$

corresponds to:

$$PC1$$

Second largest:

$$\lambda_2$$

corresponds to:

$$PC2$$

and so on.

Step 5: Select components

If we need 2 dimensions:

$$PC1,\ PC2$$

are selected.

Step 6: Project data

$$Z_{new} = ZW$$

where W contains selected eigenvectors.

17.7 Explained Variance

Explained variance tells us how much information each principal component captures.

For component i:

$$EV_i = \frac{\lambda_i}{\sum_j \lambda_j}$$

Example:

```
PC1 → 60%
PC2 → 25%
PC3 → 10%
PC4 → 5%
```

Then:

```
PC1 + PC2 = 85%
```

So two components preserve approximately 85% of the variance.

17.8 PCA Applications

- Data visualization
- Image compression
- Noise reduction
- Feature compression
- Pattern recognition
- Exploratory data analysis
- Reducing computational complexity

---

18. LDA — LINEAR DISCRIMINANT ANALYSIS

18.1 Definition

Linear Discriminant Analysis (LDA) is a supervised dimensionality reduction technique that finds projections that maximize the separation between different classes.

Main idea:

> **LDA tries to make different classes as far apart as possible while keeping samples from the same class close together.**

18.2 Example

Suppose we have two classes:

```
Class A = ●
Class B = ○
```

Initially:



LDA searches for a projection direction that separates:



18.3 LDA is Supervised

LDA requires class labels.

Example:

```
Features + Class
      ↓
     LDA
      ↓
Discriminative directions
```

This is the key difference from PCA.

18.4 Important LDA Concepts

Class Mean

Mean vector of a particular class.

Within-Class Scatter

Measures how spread out samples are **inside each class**.

Between-Class Scatter

Measures how far apart the class means are.

LDA attempts to:

```
Minimize within-class variation
+
Maximize between-class variation
```

18.5 LDA Objective

Conceptually:

$$\text{Maximize} \frac{\text{Between-Class Scatter}}{\text{Within-Class Scatter}}$$

In other words:

```
Classes far apart
       +
Samples within class close together
```

```
        ↓
Good LDA projection
```

18.6 LDA Mathematical Form

Within-class scatter:

$$S_W = \sum_{k=1}^{c} \sum_{x \in C_k} (x - \mu_k)(x - \mu_k)^T$$

Between-class scatter:

$$S_B = \sum_{k=1}^{c} N_k (\mu_k - \mu)(\mu_k - \mu)^T$$

LDA finds directions w that maximize:

$$J(w) = \frac{w^T S_B w}{w^T S_W w}$$

The resulting directions are obtained from the generalized eigenvalue problem:

$$S_B w = \lambda S_W w$$

18.7 Maximum Number of LDA Components

If there are C classes, LDA can produce at most:

$$C - 1$$

discriminant components.

Example:

```
2 classes → Maximum 1 component
3 classes → Maximum 2 components
4 classes → Maximum 3 components
```

This is a **very important exam point**.

---

19. PCA vs LDA

Feature	PCA	LDA
Full form	Principal Component Analysis	Linear Discriminant Analysis
Learning type	Unsupervised	Supervised
Labels required?	No	Yes
Main objective	Maximize variance	Maximize class separation
Uses class information?	No	Yes
Focus	Data variance	Class discrimination
Components	Up to number of original features	At most $C - 1$
Common use	Visualization/compression	Classification-related reduction
Based on	Covariance/eigenvectors	Scatter matrices/eigenvectors

⭐ Memory Trick

```
PCA → Preserve Variance

LDA → Learn Discrimination
```

---

20. FEATURE SELECTION vs FEATURE EXTRACTION vs FEATURE TRANSFORMATION vs DIMENSIONALITY REDUCTION

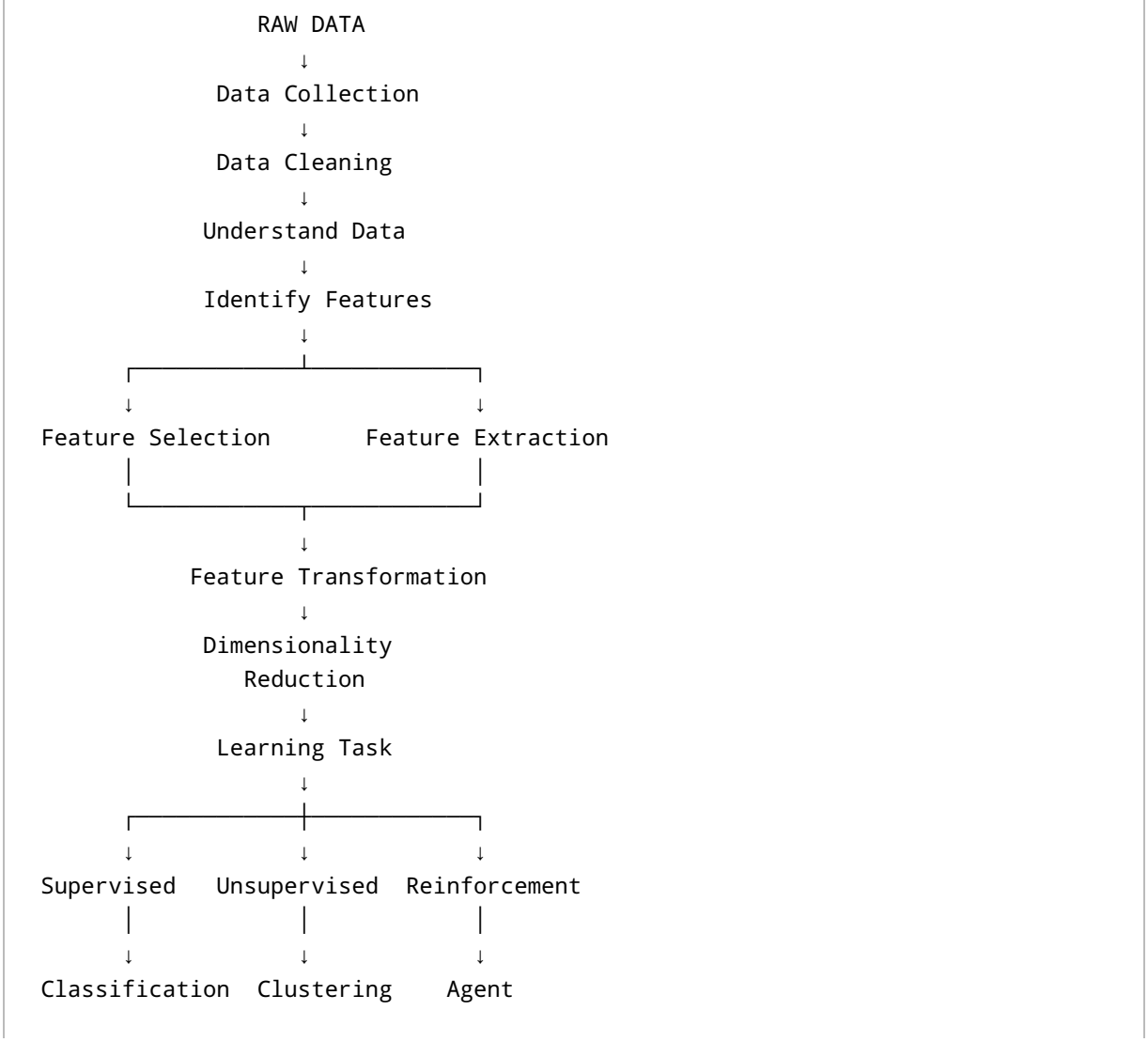
This is an extremely important conceptual comparison.

Concept	Meaning	Example
Feature Selection	Choose useful existing features	Select Age, Salary
Feature Extraction	Create useful representation from raw data	Image → edges
Feature Transformation	Change representation/scale	Standardization

Concept	Meaning	Example
Dimensionality Reduction	Reduce number of dimensions	100 → 10
PCA	Transformation + dimensionality reduction	50 → 5 PCs
LDA	Supervised projection + dimensionality reduction	20 → class-separating components

21. COMPLETE MACHINE LEARNING WORKFLOW

The concepts of Unit 1 can be combined into one ML workflow:



```
Regression      Patterns      Reward
      │
      ↓
  Train Model
      ↓
  Test Model
      ↓
  Prediction
      ↓
 Generalization
```

---

22. IMPORTANT TERMINOLOGY

Dataset

Collection of data.

Sample

One observation.

Feature

Input characteristic.

Feature Vector

Collection of feature values for one sample.

Label

Known output associated with a training example.

Target

Variable to be predicted.

Model

Learned representation of patterns.

Training

Learning from data.

Testing

Evaluating on unseen data.

Prediction

Output produced by a trained model.

Generalization

Ability to perform well on unseen data.

Dimensionality

Number of features.

Feature Selection

Selecting useful existing features.

Feature Extraction

Creating useful representations/features.

Feature Transformation

Changing feature representation or scale.

Dimensionality Reduction

Reducing the number of dimensions while preserving important information.

---

23. IMPORTANT EXAM DEFINITIONS

Machine Learning

> Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and improve their performance on a task without being explicitly programmed for every possible situation.

Supervised Learning

> Supervised learning is a learning paradigm in which a model learns a mapping between input features and known target labels from labeled training data.

Unsupervised Learning

Unsupervised learning is a learning paradigm in which algorithms discover hidden structures, patterns, or relationships in unlabeled data.

Reinforcement Learning

Reinforcement learning is a learning paradigm in which an agent learns to make decisions by interacting with an environment and receiving rewards or penalties.

Feature

A feature is an individual measurable property or characteristic used to represent an observation.

Dimensionality Reduction

Dimensionality reduction is the process of reducing the number of features while retaining the most important information present in the data.

PCA

PCA is an unsupervised dimensionality reduction technique that transforms the original features into principal components ordered according to the amount of variance they explain.

LDA

LDA is a supervised dimensionality reduction technique that finds projections that maximize separation between classes while minimizing variation within classes.

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24. VERY IMPORTANT DIFFERENCES

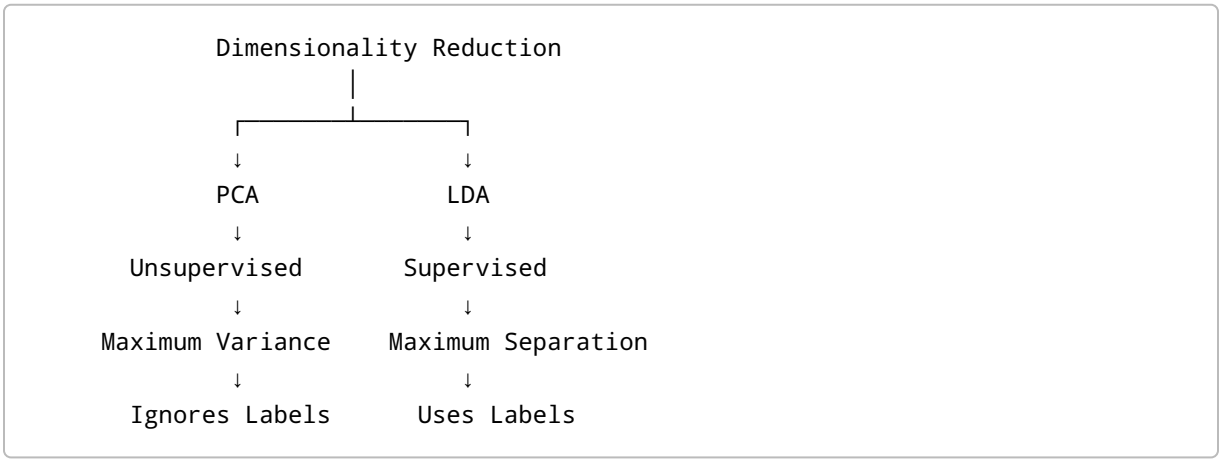
Classification vs Regression

Classification	Regression
Categorical output	Numerical output
Spam/Not Spam	House price
Pass/Fail	Salary
Cat/Dog	Temperature

Supervised vs Unsupervised

Supervised	Unsupervised
Labeled data	Unlabeled data
Target exists	Target absent
Predict output	Discover structure
Classification	Clustering
Regression	Association

PCA vs LDA



25. QUICK REVISION SHEET 🚀

```
MACHINE LEARNING
│
├── Learns patterns from DATA
│
├── DATA
│   ├── Qualitative
│   │   ├── Nominal
│   │   └── Ordinal
│   │
│   └── Quantitative
│       ├── Discrete
```

```
│       └── Continuous
│
├── SCALES
│   ├── Nominal → Name
│   ├── Ordinal → Order
│   ├── Interval → Equal interval
│   └── Ratio → True zero
│
├── DATA REPRESENTATION
│   ├── Samples
│   ├── Features
│   ├── Feature Vector
│   ├── Labels
│   └── Patterns
│
├── LEARNING TASKS
│   ├── Descriptive
│   └── Predictive
│       ├── Classification
│       └── Regression
│
├── LEARNING PARADIGMS
│   ├── Supervised → Labels
│   ├── Unsupervised → No Labels
│   └── Reinforcement → Rewards
│
├── LEARNING MODEL
│   ├── Training
│   ├── Model
│   ├── Testing
│   ├── Prediction
│   └── Generalization
│
├── FEATURES
│   ├── Feature Sets
│   ├── Feature Selection
│   ├── Feature Extraction
│   └── Feature Transformation
│
└── DIMENSIONALITY REDUCTION
    │
    ├── PCA
    │   ├── Unsupervised
    │   ├── Maximum Variance
    │   ├── Covariance
    │   ├── Eigenvectors
```

```
    │    └── Eigenvalues
    │
    └── LDA
        ├── Supervised
        ├── Class Separation
        ├── Within-Class Scatter
        ├── Between-Class Scatter
        └── Maximum Components = C − 1
```

⭐ MUST-REMEMBER POINTS

1. **ML learns patterns from data.**
2. **Feature = input variable.**
3. **Label/Target = output to be predicted.**
4. **Classification → categorical output.**
5. **Regression → numerical output.**
6. **Supervised → labeled data.**
7. **Unsupervised → unlabeled data.**
8. **Reinforcement → reward-based learning.**
9. **Nominal → categories without order.**
10. **Ordinal → categories with order.**
11. **Interval → equal intervals, no true zero.**
12. **Ratio → equal intervals + true zero.**
13. **Feature selection → choose existing features.**
14. **Feature extraction → derive/create useful features.**
15. **Feature transformation → change representation/scale.**
16. **Dimensionality reduction → fewer dimensions, retain important information.**

17. **PCA → maximize variance.**

18. **LDA → maximize class separation.**

19. **PCA does not require labels.**

20. **LDA requires labels.**

21. **PCA components are based on eigenvectors/eigenvalues of covariance structure.**

22. **LDA uses within-class and between-class scatter.**

23. **Maximum LDA components =** $C - 1$**.**

24. **Good ML models should generalize to unseen data.**

25. **Algorithm learns; model is the learned representation.**

🎯 UNIT 1 IN ONE LINE

Machine Learning learns patterns from data → represents data using features → performs descriptive/predictive tasks using different learning paradigms → builds learning models → manages features and dimensionality → and uses techniques such as PCA and LDA to obtain useful lower-dimensional representations.